

MAC-2313 | CALCULUS III

Worked Answer Key

Ordered reasoning and partial-credit checkpoints

Name

Date

Score

Eight core problems, 10 points each. One optional bonus problem is worth up to 10 additional points. Show your reasoning: a correct method with a small arithmetic error may earn partial credit.

No formula sheet is provided. Assume the single-variable derivative and integral skills from Calculus I and substitution/integration by parts from Calculus II are available when useful. Exact answers are preferred unless an estimate is requested.

Suggested practice: attempt the exam without notes, then use the separate answer key to diagnose method choices and reasoning. No time limit is specified.

Practice Exam 1

Problem 1 | Vectors and displacement

Solution:

(a) $PQ = \langle 3, 4, -2 \rangle$, $QR = \langle 3, -3, 5 \rangle$, and $PR = \langle 6, 1, 3 \rangle$; indeed $PQ + QR = PR$. (b) The traveled length is $|PQ| + |QR| = \sqrt{29} + \sqrt{43}$ m. The endpoint distance is $|PR| = \sqrt{46}$ m. Path length accumulates both legs; displacement depends only on endpoints. (c) $\langle 6, 1, 3 \rangle / \sqrt{46}$. (d) It does not determine the intermediate waypoint or the traveled path length.

Partial-credit guidance:

Partial credit (10): 3 for leg vectors; 2 for net vector; 2 for both lengths; 1 for explaining route vs endpoint; 1 for unit vector; 1 for a valid limitation.

Problem 2 | Dot products, projection, and work

Solution:

$|d|=13$, $|F|=10$, and $F \cdot d = 18 + 32 = 50$, so $W = 50$ J (positive: force has a component along the motion). The scalar component is $(F \cdot d)/|d| = 50/13$ N. The vector projection is $((F \cdot d)/(d \cdot d))d = (50/169)\langle 3, 4, 12 \rangle$ N. Finally $\cos(\theta) = 50/(10 \cdot 13) = 5/13$. Work multiplies the along-motion force component by the displacement length; the projection is force, not energy.

Partial-credit guidance:

Partial credit (10): 2 for dot product/work and sign; 2 for scalar projection; 3 for vector projection; 1 for cosine; 2 for distinguishing units and meanings.

Problem 3 | Cross products and plane equations

Solution:

$AB = \langle 2, -1, 1 \rangle$, $AC = \langle -1, 2, 2 \rangle$. $AB \times AC = \langle -4, -5, 3 \rangle$, so a normal is $\langle 4, 5, -3 \rangle$. Through A: $4(x-1) + 5(y-2) - 3z = 0$, or $4x + 5y - 3z = 14$. The parallelogram area is $|AB \times AC| = \sqrt{16 + 25 + 9} = 5\sqrt{2}$. At D, $4(2) + 5(2) - 3(4) = 6$, not 14, so D is not on the plane. Reversing the order negates the normal and orientation, but preserves its magnitude and the plane it determines.

Partial-credit guidance:

Partial credit (10): 2 for vectors; 2 for cross-product normal; 2 for plane; 1 for area; 1 for membership test; 2 for order interpretation.

Problem 4 | First and second partial derivatives

Solution:

Holding y fixed: $H_x = 4xy - y^2 + 3$. Holding x fixed: $H_y = 2x^2 - 2xy + 5$. Thus at $(1, 2)$, $H_x = 7$ and $H_y = 3$. Differentiate again: $H_{xx} = 4y = 8$, $H_{xy} = 4x - 2y = 0$, $H_{yx} = 4x - 2y = 0$, $H_{yy} = -2x = -2$. The first-order change is $dH \approx H_x dx + H_y dy = 7(0.02) + 3(-0.01) = 0.11$ response units. The mixed partials agree here.

Partial-credit guidance:

Partial credit (10): 2 for first partials; 4 for second/mixed partials; 2 for holding the other variable fixed; 2 for differential estimate.

Problem 5 | Differentials and approximation

Solution:

$S_r = 4\pi r + 2\pi h$ and $S_h = 2\pi r$. At $(3, 10)$, these are 32π and 6π . Therefore $dS \approx 32\pi(0.02) + 6\pi(-0.05) = 0.34\pi$ $\text{cm}^2 \approx 1.068 \text{ cm}^2$. Since $S = 78\pi$, estimated relative change is $(0.34\pi)/(78\pi) \cdot 100\% \approx 0.436\%$. It rises because the radius increase effect exceeds the height decrease effect. The differential omits higher-order terms in dr and dh .

Partial-credit guidance:

Partial credit (10): 3 for partials; 2 for substitution and dS ; 2 for relative percent; 1 for direction; 2 for explaining linearization.

Problem 6 | Multivariable chain rule

Solution:

At $t=1$, $(x,y)=(2,1)$, $x'=2$, $y'=-1$. $C_x=2xe^y+y \cos(xy)$, $C_y=x^2e^y+x \cos(xy)$, so at $(2,1)$ they are $4e+\cos 2$ and $4e+2\cos 2$. Thus $dC/dt=2(4e+\cos 2)-(4e+2\cos 2)=4e$. Direct check: $x^2e^y=(1+t^2)^2e^{(2-t)}$, whose derivative at 1 is $e(8-4)=4e$. Also $xy=(1+t^2)(2-t)$, whose derivative at 1 is $2-2=0$, so $d/dt[\sin(xy)]=\cos(2)\cdot 0=0$. Both routes give $4e$.

Partial-credit guidance:

Partial credit (10): 2 for inner values/rates; 3 for correct partials; 2 for chain combination; 2 for direct check; 1 for identifying cancellation in the sinusoidal term.

Problem 7 | Gradient and directional derivative

Solution:

$\nabla Z = \langle -2x-2y, -2x-4y \rangle$, so $\nabla Z(1,-1) = \langle 0, 2 \rangle$. The direction $P \rightarrow Q$ is $\langle 3, 4 \rangle$, hence $u = \langle 3/5, 4/5 \rangle$. $D_u Z = \nabla Z \cdot u = 8/5$ elevation units per horizontal unit, so elevation rises in that direction. Steepest increase points along $\nabla Z / |\nabla Z| = \langle 0, 1 \rangle$ at rate $|\nabla Z| = 2$. Steepest decrease is $\langle 0, -1 \rangle$ at rate -2 .

Partial-credit guidance:

Partial credit (10): 3 for gradient; 2 for normalizing the requested direction; 2 for directional rate and interpretation; 2 for steepest increase; 1 for steepest decrease.

Problem 8 | Critical points and optimization

Solution:

$P_x = 32 - 4x - 2y$; $P_y = 36 - 2x - 6y$. Setting both to zero gives $2x + y = 16$ and $x + 3y = 18$, hence $(x, y) = (6, 4)$, feasible. Hessian entries are $P_{xx} = -4$, $P_{xy} = -2$, $P_{yy} = -6$; $D = 24 - 4 = 20 > 0$ and $P_{xx} < 0$, so strict local maximum. The Hessian is negative definite everywhere (leading minor $-4 < 0$, determinant $20 > 0$), so P is strictly concave and this critical point is the unique global maximum on the convex quadrant. $P(6, 4) = 168$, i.e. \$16,800 at 600 and 400 items in the model's units.

Partial-credit guidance:

Partial credit (10): 3 for solving gradient equations; 2 for second derivative test; 2 for global argument; 2 for value; 1 for units/context.

Optional bonus — variable-force work | Bonus

Solution:

$W = \int_1^2 \mathbf{F} \cdot d\mathbf{r} = \int_1^2 x e^x dx$. Integration by parts with $u=x$, $dv=e^x dx$ gives $du=dx$, $v=e^x$, so

$W = [x e^x]_1^2 - \int_1^2 e^x dx = 2e^2 - (e^2 - 1) = e^2 + 1$ J. A single constant-force dot product does not apply because $\mathbf{F}$ changes along the path.

Partial-credit guidance:

Partial credit (10): 2 for work integral; 3 for integration-by-parts setup; 3 for evaluation; 2 for explaining variable force.

Practice Exam 2

Problem 1 | Vectors and displacement

Solution:

$\vec{AB} = \langle 3, -3, 3 \rangle$; $\vec{BC} = \langle -4, 4, 2 \rangle$; $\vec{AC} = \langle -1, 1, 5 \rangle$. Their sum is $\vec{AC}$. $|\vec{AB}| = 3\sqrt{3}$, $|\vec{BC}| = 6$, $|\vec{AC}| = 3\sqrt{3}$. Path length $A \rightarrow B \rightarrow C$ is $3\sqrt{3} + 6$, while endpoint distance is $3\sqrt{3}$. The unit net direction is $\langle -1, 1, 5 \rangle / (3\sqrt{3})$. Returning $C \rightarrow A$ has displacement $\langle 1, -1, -5 \rangle$ and length $3\sqrt{3}$, so the closed-trip distance is $6\sqrt{3} + 6$. Net displacement is zero for the full loop, while distance is positive because motion occurred.

Partial-credit guidance:

Partial credit (10): 3 for displacements; 2 for length/distance; 1 for unit direction; 2 for return vector and distance; 2 for interpretation.

Problem 2 | Dot products, projection, and work

Solution:

$\mathbf{F} \cdot \mathbf{d} = -4 - 5 + 8 = -1$, so $W = -1$ J: the net component along motion opposes it. $|\mathbf{d}| = 3$ and $\mathbf{d} \cdot \mathbf{d} = 9$. Scalar projection is $(\mathbf{F} \cdot \mathbf{d})/|\mathbf{d}| = -1/3$ N; vector projection is $((\mathbf{F} \cdot \mathbf{d})/(\mathbf{d} \cdot \mathbf{d}))\mathbf{d} = (-1/9)\langle 2, -1, 2 \rangle = \langle -2/9, 1/9, -2/9 \rangle$ N. $|\mathbf{F}| = \sqrt{45} = 3\sqrt{5}$, so $\cos(\theta) = -1/(9\sqrt{5}) < 0$ and θ is obtuse. Nonzero force can have an opposing component, which removes energy from motion.

Partial-credit guidance:

Partial credit (10): 2 for work/sign; 2 scalar projection; 3 vector projection; 1 cosine/angle; 2 interpretation.

Problem 3 | Cross products and plane equations

Solution:

$AB = \langle 2, -1, -1 \rangle$, $AC = \langle 1, 2, -2 \rangle$. $AB \times AC = \langle 4, 3, 5 \rangle$, a normal. Through A gives $4x + 3(y - 1) + 5(z - 2) = 0$, or $4x + 3y + 5z = 13$. The parallelogram area is $\sqrt{16 + 9 + 25} = 5\sqrt{2}$; triangle area is half, $5\sqrt{2}/2$. At D, $4 + 3 + 5 = 12 \neq 13$, so D is outside the sheet. If the cross product were zero, the direction vectors would be parallel and the points collinear, so they would not define a unique plane.

Partial-credit guidance:

Partial credit (10): 2 vectors; 2 cross product; 2 plane; 2 areas; 1 membership; 1 collinearity criterion.

Problem 4 | First and second partial derivatives

Solution:

$f_s = e^s \sin t + t^2$; $f_t = e^s \cos t + 2st$. Second derivatives: $f_{ss} = e^s \sin t$; $f_{st} = e^s \cos t + 2t$; $f_{ts} = e^s \cos t + 2t$; $f_{tt} = -e^s \sin t + 2s$. At $(0, \pi/2)$: $f_s = 1 + \pi^2/4$, $f_t = 0$, $f_{ss} = 1$, $f_{st} = f_{ts} = \pi$, $f_{tt} = -1$. Thus $df \approx f_s ds + f_t dt = (1 + \pi^2/4)(0.01) \approx 0.0347$ response units. The dt term is zero to first order at the specified point.

Partial-credit guidance:

Partial credit (10): 2 first partials; 4 second/mixed partials; 2 point evaluation; 2 differential estimate.

Problem 5 | Differentials and approximation

Solution:

$S_r = 4\pi r + 2\pi h = 30\pi$ and $S_h = 2\pi r = 8\pi$ at the given point. $dS \approx 30\pi(-0.03) + 8\pi(0.04) = -0.58\pi \text{ cm}^2 \approx -1.822 \text{ cm}^2$. Since $S = 88\pi$, relative change $\approx -0.58/88 \cdot 100\% = -0.659\%$. The radius reduction dominates because its weighted contribution (-0.90π) exceeds the height contribution $(+0.32\pi)$. Area depends jointly and nonlinearly on r, h , so combine partial-derivative contributions; multiplying dimension percentages ignores the model's sensitivities and cross terms.

Partial-credit guidance:

Partial credit (10): 3 for partials; 2 for dS ; 2 relative change; 1 sign; 2 sensitivity explanation.

Problem 6 | Multivariable chain rule

Solution:

At $t=1$, $x=y=z=2$ and $(x',y',z')=(2,-1,2)$. $T_x=2xy=8$, $T_y=x^2+z=6$, $T_z=y=2$. Therefore $dT/dt=8(2)+6(-1)+2(2)=14$.
Direct substitution: $T=(t^2+1)^2(3-t)+2t(3-t)$. Differentiating and evaluating at 1 gives 14, matching the chain rule.
The three paths contribute 16, -6, and 4.

Partial-credit guidance:

Partial credit (10): 2 point/rates; 3 partial derivatives; 2 chain sum; 2 direct check; 1 interpretation of paths.

Problem 7 | Gradient and directional derivative

Solution:

$\nabla c = \langle 6x - 2y, 2y - 2x \rangle$, so $\nabla c(1, 1) = \langle 4, 0 \rangle$. The direction toward Q is $\langle -3, 4 \rangle$, giving unit vector $u = \langle -3/5, 4/5 \rangle$.

Directional derivative is $\nabla c \cdot u = -12/5$ units per distance, so cost locally decreases toward Q. Fastest increase is $\langle 1, 0 \rangle$ at rate 4; fastest decrease is $\langle -1, 0 \rangle$ at rate -4. The target direction is not a descent direction because it has a negative dot product with the gradient.

Partial-credit guidance:

Partial credit (10): 3 gradient; 2 normalized direction; 2 directional rate/interpretation; 2 steepest directions; 1 max/min rates.

Problem 8 | Constrained optimization

Solution:

$Px = 22 - 2x - 2y$ and $Py = 35 - 2x - 6y$. Solving gives $2x + 2y = 22$ and $2x + 6y = 35$, so $(x, y) = (31/4, 13/4)$, whose sum is 11; it is infeasible. On $y = 10 - x$, $0 \leq x \leq 10$: $P = -2x^2 + 27x + 50$. The vertex is $x = 27/4$, $y = 13/4$, with $P = 1129/8 = 141.125$. Other edges: $y = 0$ gives $22x - x^2$, max 120 at $x = 10$; $x = 0$ gives $35y - 3y^2$, max $1225/12 \approx 102.083$ at $y = 35/6$. Capacity-edge endpoints have values 50 and 50. Thus the constrained maximum is at $(27/4, 13/4)$, with modeled profit \$14,112.50. Unconstrained calculus ignores feasibility and can select a prohibited production plan.

Partial-credit guidance:

Partial credit (10): 2 unconstrained solve/feasibility; 3 edge reduction/vertex; 2 other boundaries; 2 compare/classify maximum; 1 contextual value.

Optional bonus — variable-force work | Bonus

Solution:

$W = \int_0^{\pi/2} \sin(x)\cos(x)dx$. Let $u = \sin x$, $du = \cos x dx$; limits change from 0 to 1. Then $W = \int_0^1 u du = 1/2$ J. The force varies with position, so work accumulates as the along-path force component changes; a single constant-force dot product is not valid.

Partial-credit guidance:

Partial credit (10): 2 work integral; 3 substitution and limits; 3 evaluation; 2 explanation.

Source and scope notes

The topic scope and format follow the supplied MAC-2313 Midterm Exam Details and Study Guide. Method alignment was checked against the locally imported OpenStax Calculus, Volume 3 (Chapter 2 and Sections 4.3–4.7) and the existing Stewart Calculus: Early Transcendentals, Sixth Edition source (Sections 12.2–12.5 and 14.3–14.7). The two optional work problems use standard substitution and integration by parts (Stewart §§5.5, 7.1). Problems and solutions here are original; no textbook exercise wording is reproduced.

Suggested point breakdowns support self-review and are not a substitute for an instructor's grading rubric.